Understanding generated files in OpenLane

INTRODUCTION:

In this lab we will be understanding generated files in OpenLane, this lab provides comprehensive insight into the RTL-to-GDSII design flow by analysing the automated reports generated across synthesis, floorplan, placement, clock tree synthesis, routing, timing analysis, power estimation, and final signoff verification stages. These structured log files detail key metrics like timing slack, power consumption, area utilization, routing congestion, DRC/LVS violations, all accessible in the folder.

APPLICATION:

The reports are essential for verifying design quality and making optimization decisions throughout the RTL-to-GDSII flow. We use synthesis reports to check cell count and area after Yosys, floorplan reports to validate core utilization and IO placement, placement reports to confirm density targets, and CTS reports to measure clock skew. Routing reports reveal congestion and wirelength issues, while STA reports identify timing violations for config adjustments. Power reports quantify dynamic/leakage consumption, and DRC/LVS reports confirm manufacturing readiness. Engineers compare these metrics across PDK runs, track iteration improvements, and compile final signoff packages proving tapeout readiness to foundries.

OBJECTIVE:

After completing this lab, you will be able to:

• Understand the role of reports generated after every flow

• Understand stat reports after synthesis

• Understand die area and core area reports after floorplan

• Understand reports of global placement and detailed placement after placement

• Understand timing reports after the process clock tree synthesis

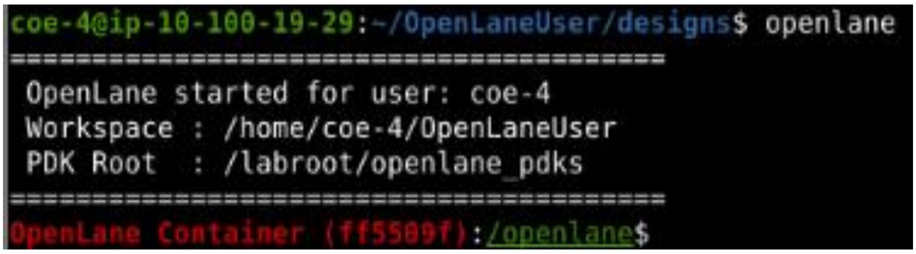
• Understand reports of global routing and detailed routing after routing

• Understand DRV/LVS reports after signoff

PROCEDURE TO CREATE A LAB:

Step 1: Set up OpenLane flow

1.1: Invoke OpenLane in the terminal using the command: `openlane`



1.2: Open folder- designs: `cd designs`

Step 2: Set up Project Directory

2.1: Create a directory for your lab: mkdir <project_name>

2.2: Open lab directory: cd <project_name>

2.3: Create directory for project: `mkdir fd`

2.4: Open project directory: `cd fd`

2.5: Inside project directory, create folder: `mkdir src`

2.6: Open folder: `cd src`

Step 3: Create Design file

3.1: Open vi editor for design with command: `vi fd.v`

RTL Design Code:

```
module fd(
        input clk,
        input reset,
        output clk_out);
        reg[1:0] pos_cnt;
        reg [1:0]neg_cnt;

        always@(posedge clk)
                if(reset)
                        pos_cnt<= 0;
                else if(pos_cnt==2)
```

```
                        pos_cnt<= 0;
                else
                        pos_cnt<= pos_cnt+1;

        always@(negedge clk)
        if(reset)
                neg_cnt<= 0;
        else if(neg_cnt==2)
                neg_cnt<= 0;
        else neg_cnt<= neg_cnt+1;

        assign clk_out=((pos_cnt==2) | (neg_cnt==2));

endmodule
```

3.2: save the file using command- :`wq`

Step 4: Create .json file

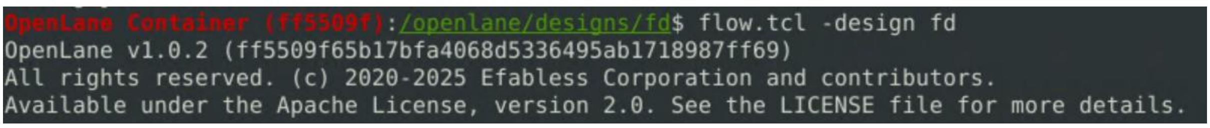
4.1: Open vi editor: `vi config.json`

Note: Follow the previous lecture for creating config.json file.

Step 5: Use the following commands to run the flow:

OpenLane to start the design flow in one go.

```
flow.tcl -design fd
```



After completion of flow, checks for generated reports:

```
coe-2@ip-10-100-19-29:~/OpenLaneUser/designs/fd$ ls
config.json  runs  src
```

Path for Results:

/OpenLaneUser/designs/fd/runs/RUN_2026.01.30_07.29.02/reports

Step 6: To see results folder

6.1: Open result directory: `cd reports`

```
coe-2@ip-10-100-19-29:~/OpenLaneUser/designs/fd/runs/RUN_2026.01.30_07.29.02$ cd reports/
coe-2@ip-10-100-19-29:~/OpenLaneUser/designs/fd/runs/RUN_2026.01.30_07.29.02/reports$ ls
cts  floorplan  manufacturability.rpt  metrics.csv  placement  routing  signoff  synthesis
```

6.2: Open synthesis folder: `cd synthesis`

```
coe-2@ip-10-100-19-29:~/OpenLaneUser/designs/fd/runs/RUN_2026.01.30_07.29.02/reports$ cd synthesis/
coe-2@ip-10-100-19-29:~/OpenLaneUser/designs/fd/runs/RUN_2026.01.30_07.29.02/reports/synthesis$
```

Do, `ls`

These are the different reports we will be able to observe after synthesis

```
coe-2@ip-10-100-19-29:~/OpenLaneUser/designs/fd/runs/RUN_2026.01.30_07.29.02/reports/synthesis$ ls
1-synthesis.AREA_0.chk.rpt   1-synthesis_pre.stat            2-syn_sta.max.rpt    2-syn_sta.skew.rpt
1-synthesis.AREA_0.stat.rpt  1-synthesis_pre_synth.chk.rpt   2-syn_sta.min.rpt    2-syn_sta.summary.rpt
1-synthesis_dff.stat         2-syn_sta.checks.rpt            2-syn_sta.power.rpt
```

6.2.1: Open folder 1-synthesis_dff.stat : `vi 1-synthesis_dff.stat`

```
53. Printing statistics.

=== fd ===

   Number of wires:                 19
   Number of wire bits:             25
   Number of public wires:           5
   Number of public wire bits:       7
   Number of memories:               0
   Number of memory bits:            0
   Number of processes:              0
   Number of cells:                 19
     $_ANDNOT_                       2
     $_MUX_                          4
     $_NOT_                          4
     $_OR_                           3
     $_XOR_                          2
     sky130_fd_sc_hd__dfxtp_2        4

~
~
~
```

Note: We can open all reports and observe

6.3: Open floorplan folder: `cd floorplan`

```
coe-2@ip-10-100-19-29:~/OpenLaneUser/designs/fd/runs/RUN_2026.01.30_07.29.02/reports$ cd floorplan/
```

Do, `ls`

These are the different reports we will be able to observe after floorplan

```
coe-2@ip-10-100-19-29:~/OpenLaneUser/designs/fd/runs/RUN_2026.01.30_07.29.02/reports/floorplan$ ls
3-initial_fp_core_area.rpt  3-initial_fp_die_area.rpt
```

6.3.1: Open folder 3-initial_fp_core_area.rpt : `vi 3-initial_fp_core_area.rpt`

```
5.52 10.88 28.98 46.24
~
~
~
```

Note: We can open all reports and observe

6.4: Open floorplan folder: `cd placement`

```
coe-2@ip-10-100-19-29:~/OpenLaneUser/designs/fd/runs/RUN_2026.01.30_07.29.02/reports$ cd placement/
```

Do, `ls`

These are the different reports we will be able to observe after placement

```
coe-2@ip-10-100-19-29:~/OpenLaneUser/designs/fd/runs/RUN_2026.01.30_07.29.02/reports/placement$ ls
10-dpl_sta.checks.rpt  10-dpl_sta.max.rpt  10-dpl_sta.min.rpt  10-dpl_sta.power.rpt  10-dpl_sta.skew.rpt  10-dpl_sta.summary.rpt
```

6.4.1: Open folder 10-dpl_sta.max.rpt: `vi 10-dpl_sta.checks.rpt`

```
===========================================================================
report_checks -unconstrained
===========================================================================
======================= Typical Corner ===================================

Startpoint: reset (input port clocked by __VIRTUAL_CLK__)
Endpoint: _19_ (rising edge-triggered flip-flop)
Path Group: unconstrained
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                           2.00    2.00 v input external delay
                   0.01    0.00    2.00 v reset (in)
     1    0.00                           reset (net)
                   0.01    0.00    2.00 v input2/A (sky130_fd_sc_hd__buf_1)
                   0.08    0.12    2.12 v input2/X (sky130_fd_sc_hd__buf_1)
     4    0.01                           net2 (net)
                   0.08    0.00    2.12 v _10_/A (sky130_fd_sc_hd__nor3_1)
                   0.17    0.22    2.34 ^ _10_/Y (sky130_fd_sc_hd__nor3_1)
     1    0.00                           _04_ (net)
                   0.17    0.00    2.34 ^ _19_/D (sky130_fd_sc_hd__dfxtp_1)
                                   2.34   data arrival time
-----------------------------------------------------------------------------
(Path is unconstrained)
```

Note: We can open all reports and observe

6.5: Open floorplan folder: `cd routing`

```
OpenLane Container (ff5509f):/openlane/designs/fd/runs/RUN_2026.01.30_10.03.59/reports$ cd routing/
```

Do, `ls`

These are the different reports we will be able to observe after Routing

6.5.1: Open folder 1-synthesis_dff.stat : `vi 13-rsz_design_sta.power.rpt`

```
===========================================================================
 report_power
===========================================================================
======================= Typical Corner ===================================

Group                  Internal  Switching    Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------
Sequential             2.02e-06   7.60e-07   3.38e-11   2.78e-06  62.7%
Combinational          8.77e-07   7.78e-07   1.31e-10   1.66e-06  37.3%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
----------------------------------------------------------------
Total                  2.90e-06   1.54e-06   1.65e-10   4.44e-06 100.0%
                          65.3%      34.7%       0.0%
```

Note: We can open all reports and observe

6.6: Open floorplan folder: `cd signoff`

```
OpenLane Container (ff5509f):/openlane/designs/fd/runs/RUN_2026.01.30_10.03.59/reports$ cd signoff/
```

Do, `ls`

These are the different reports we will be able to observe after signoff

```
OpenLane Container (ff5509f):/openlane/designs/fd/runs/RUN_2026.01.30_10.03.59/reports/signoff$ ls
23-mca                28-rcx_sta.min.rpt      29-irdrop-VPWR.rpt            38-antenna_violators.rpt  drc.tr
25-mca                28-rcx_sta.power.rpt    32-xor.rpt                    drc.klayout.xml           fd.rpt
27-mca                28-rcx_sta.skew.rpt     32-xor.xml                    drc.rdb                   fd.rpt.debug.gz
28-rcx_sta.checks.rpt 28-rcx_sta.summary.rpt  36-fd.lvs.rpt                 drc.rpt                   fd.rpt.error.gz
28-rcx_sta.max.rpt    29-irdrop-VGND.rpt      38-antenna_violators_pins.txt drc.tcl                   spice.feedback.txt
```

6.6.1: Open folder 36-fd.lvs.rpt : `vi 36-fd.lvs.rpt`

```
LVS reports no net, device, pin, or property mismatches.

Total errors = 0
~
~
~
```

Note: We can open all reports and observe

OBSERVATION:

After completion of this lab, we will be able to observe

- Synthesis showed cell count before floorplan
- Floorplan reports confirmed exact die area match and core utilization
- Detailed placement and global placement achieved target
- Global routing and detailed routing finished with zero DRC violations

CONCLUSION:

In this OpenLane Reports Lab, all automated RTL-to-GDSII flow reports were successfully generated and analysed for the given design. Synthesis reports confirmed cell count and initial timing, floorplan reports validated die/core areas and utilization targets, placement reports showed global-to-detailed density optimization, CTS reports verified low clock skew, routing reports confirmed congestion-free completion, and final signoff reports achieved clean DRC/LVS results.